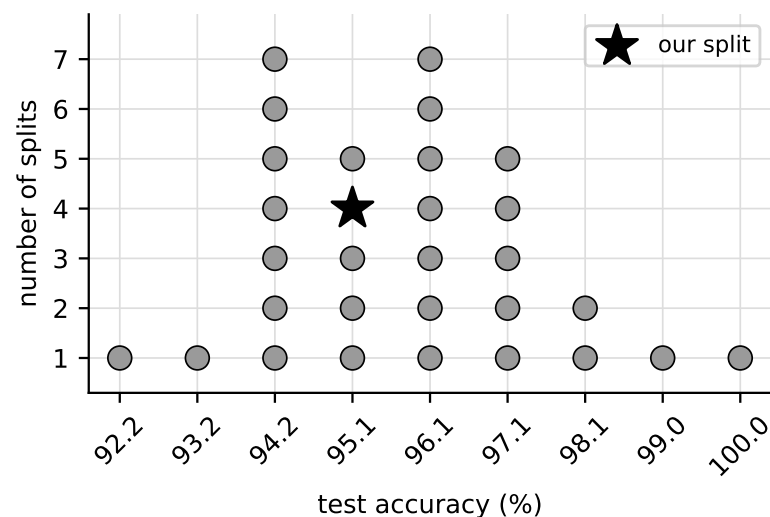


DATA 202 — Testing a Model · Week 6 Wednesday

Same penguins as today's slides: 342 penguins, split at random into 239 for training and 103 for testing. Your professor will tell you when to do each part.

1 · 30 splits, 30 scores



1. Each dot is one random split: same model ($k = 5$), same penguins. ★ = our split.
2. Lowest score: _____ %
Highest score: _____ %
3. **Circle** the split you'd be tempted to report.

2 · Build the confusion matrix

Our split, $k = 5$. The model got **46** Adelies, **18** Chinstraps and **34** Gentoos right. Its 5 mistakes:

bill length (mm)	bill depth (mm)	true species	predicted
44.4	17.3	Gentoo	Chinstrap
45.8	18.9	Adelie	Chinstrap
48.1	16.4	Chinstrap	Gentoo
40.9	16.6	Chinstrap	Adelie
42.9	17.6	Adelie	Chinstrap

1. Fill in all nine cells. Rows = **true species**, columns = **predicted**.

	Adelie	Chinstrap	Gentoo
Adelie	_____	_____	_____
Chinstrap	_____	_____	_____
Gentoo	_____	_____	_____

2. Accuracy = right $\div$ all = _____ $\div$ 103 = _____

3 · Finding the Chinstraps

1. In your matrix, **circle** the Chinstrap row and **box** the Chinstrap column.

2. **Recall** — of the real Chinstraps, how many did we find?

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

3. **Precision** — of the penguins we called Chinstrap, how many really are?

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

4. A lazy model answers *not Chinstrap* for every penguin. On “Chinstrap or not?”:

$$\text{accuracy} = \frac{\text{TN} + \text{FP}}{\text{TN} + \text{FP} + \text{TP} + \text{FN}} = \frac{\text{TN} + \text{FP}}{\text{TN} + \text{FP} + \text{TP} + \text{FN}} \cdot \text{recall} = \frac{\text{TN} + \text{FP}}{\text{TN} + \text{FP} + \text{TP} + \text{FN}}$$

5. In your matrix, write **FP** in each false-positive cell and **FN** in each false-negative cell for Chinstrap.